

In the third step we now extend our homeomorphism $\varphi: \tilde{H} \rightarrow \tilde{H}'$ to all of S^2 . This can be done analogously to the second step, face by face. By Proposition 4.2.6, all face boundaries in $\tilde{H}$ and $\tilde{H}'$ are cycles. Now if f is a face of $\tilde{H}$ and C its boundary, then $\tilde{\sigma}(C) := \bigcup \{ \tilde{\sigma}(e) \mid e \in E(C) \}$ bounds the face $\tilde{\sigma}(f)$ of $\tilde{H}'$. By Theorem 4.1.4, we may therefore extend the homeomorphism $\varphi: C \rightarrow \tilde{\sigma}(C)$ defined so far to a homeomorphism from $C \cup f$ to $\tilde{\sigma}(C) \cup \tilde{\sigma}(f)$. We finally take the union of all these homeomorphisms, one for every face f of $\tilde{H}$, as our desired homeomorphism $\varphi: S^2 \rightarrow S^2$; as before, continuity is easily checked. $\square$

Let us return now to our original goal, the definition of equivalence for planar embeddings. Let us call two planar embeddings ϱ, ϱ' of a graph G *topologically* (respectively, *combinatorially*) *equivalent* if $\varrho' \circ \varrho^{-1}$ is a topological (respectively, combinatorial) isomorphism between $\varrho(G)$ and $\varrho'(G)$. If G is 2-connected, the two definitions coincide by Theorem 4.3.1, and we simply speak of *equivalent* embeddings. Clearly, this is indeed an equivalence relation on the set of planar embeddings of any given graph.

Note that two drawings of G resulting from inequivalent embeddings may well be topologically isomorphic (exercise): for the equivalence of two embeddings we ask not only that some (topological or combinatorial) isomorphism exist between the their images, but that the canonical isomorphism $\varrho' \circ \varrho^{-1}$ be a topological or combinatorial one.

Even in this strong sense, 3-connected graphs have only one embedding up to equivalence:

[12.7.4] **Theorem 4.3.2.** (Whitney 1933)
Any two planar embeddings of a 3-connected graph are equivalent.

(4.2.7) *Proof.* Let G be a 3-connected graph with planar embeddings $\varrho: G \rightarrow H$ and $\varrho': G \rightarrow H'$. By Theorem 4.3.1 it suffices to show that $\varrho' \circ \varrho^{-1}$ is a graph-theoretical isomorphism, i.e. that $\varrho(C)$ bounds a face of H if and only if $\varrho'(C)$ bounds a face of H' , for every subgraph $C \subseteq G$. This follows at once from Proposition 4.2.7. $\square$

4.4 Planar graphs: Kuratowski's theorem

planar A graph is called *planar* if it can be embedded in the plane: if it is isomorphic to a plane graph. A planar graph is *maximal*, or *maximally planar*, if it is planar but cannot be extended to a larger planar graph by adding an edge (but no vertex).

Drawings of maximal planar graphs are clearly maximally plane. The converse, however, is not obvious: when we start to draw a planar graph, could it happen that we get stuck half-way with a proper subgraph that is already maximally plane? Our first proposition says that

this can never happen, that is, a plane graph is never maximally plane just because it is badly drawn:

Proposition 4.4.1.

- (i) Every maximal plane graph is maximally planar.
- (ii) A planar graph with $n \geq 3$ vertices is maximally planar if and only if it has $3n - 6$ edges.

Proof. Apply Proposition 4.2.8 and Corollary 4.2.10. □

(4.2.8)
(4.2.10)

Which graphs are planar? As we saw in Corollary 4.2.11, no planar graph contains K^5 or $K_{3,3}$ as a topological minor. Our aim in this section is to prove the surprising converse, a classic theorem of Kuratowski: any graph without a topological K^5 or $K_{3,3}$ minor is planar.

Before we prove Kuratowski's theorem, let us note that it suffices to consider ordinary minors rather than topological ones:

Lemma 4.4.2. A graph contains K^5 or $K_{3,3}$ as a minor if and only if it contains K^5 or $K_{3,3}$ as a topological minor.

Proof. By Proposition 1.7.3 it suffices to show that every graph G with a K^5 minor contains either K^5 as a topological minor or $K_{3,3}$ as a minor. So suppose that $G \succcurlyeq K^5$, and let K be a minimal model of K^5 in G . Then every branch set of K induces a tree in K , and between any two branch sets K has exactly one edge. If we take the tree induced by a branch set V_x and add to it the four edges joining it to other branch sets, we obtain another tree, T_x say. By the minimality of K , the tree T_x has exactly 4 leaves, the 4 neighbours of V_x in other branch sets (Fig. 4.4.1).

(1.7.3)

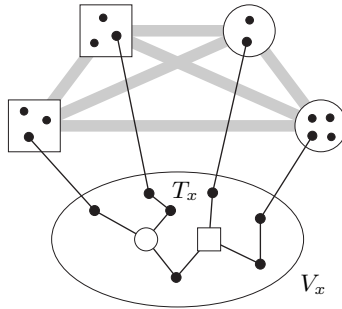


Fig. 4.4.1. Every IK^5 contains a TK^5 or $IK_{3,3}$

If each of the five trees T_x is a $TK_{1,4}$ then K is a TK^5 , and we are done. If one of the T_x is not a $TK_{1,4}$ then it has exactly two vertices of degree 3. Contracting V_x onto these two vertices, and every other branch set to a single vertex, we obtain a graph on 6 vertices containing a $K_{3,3}$. Thus, $G \succcurlyeq K_{3,3}$ as desired. □

We first prove Kuratowski's theorem for 3-connected graphs. This is the heart of the proof: the general case will then follow easily.

Lemma 4.4.3. *Every 3-connected graph G without a K^5 or $K_{3,3}$ minor is planar.*

Proof. We apply induction on $|G|$. For $|G| = 4$ we have $G = K^4$, and the assertion holds. Now let $|G| > 4$, and assume the assertion is true for smaller graphs. By Lemma 3.2.4, G has an edge xy such that G/xy is again 3-connected. Since the minor relation is transitive, G/xy has no K^5 or $K_{3,3}$ minor either. Thus, by the induction hypothesis, G/xy has a drawing $\tilde{G}$ in the plane. Let f be the face of $\tilde{G} - v_{xy}$ containing the point v_{xy} , and let C be the boundary of f . Let $X := N_G(x) \setminus \{y\}$ and $Y := N_G(y) \setminus \{x\}$; then $X \cup Y \subseteq V(C)$, because $v_{xy} \in f$. Clearly,

$$\tilde{G}' := \tilde{G} - \{v_{xy}v \mid v \in Y \setminus X\}$$

may be viewed as a drawing of $G - y$, in which the vertex x is represented by the point v_{xy} (Fig. 4.4.2). Our aim is to add y to this drawing to obtain a drawing of G .

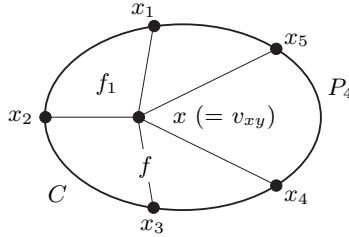


Fig. 4.4.2. $\tilde{G}'$ as a drawing of $G - y$: the vertex x is represented by the point v_{xy}

Since $\tilde{G}$ is 3-connected, $\tilde{G} - v_{xy}$ is 2-connected, so C is a cycle (Proposition 4.2.6). Let $x_1, \dots, x_k$ be an enumeration along this cycle of the vertices in X , and let $P_i = x_i \dots x_{i+1}$ be the X -paths on C between them ($i = 1, \dots, k$; with $x_{k+1} := x_1$). Let us show that $Y \subseteq V(P_i)$ for some i . Suppose not. If y has a neighbour $y' \in \dot{P}_i$ for some i , it has another neighbour $y'' \in C - P_i$, and these are separated in C by $x' := x_i$ and $x'' := x_{i+1}$. If $Y \subseteq X$ and $|Y \cap X| \leq 2$, then y has exactly two neighbours y', y'' on C but not in the same P_i , so again y' and y'' are separated on C by two vertices $x', x'' \in X$. In either case, x, y', y'' and y, x', x'' are the branch vertices of a $TK_{3,3}$ in G , a contradiction. The only remaining case is that y and x have three common neighbours on C . Then these form a TK^5 with x and y , again a contradiction.

Fix i so that $Y \subseteq P_i$. The point set $C \setminus P_i$ is contained in one of the two faces of the cycle $C_i := xx_iP_ix_{i+1}x$; we denote the *other* face of

C_i by f_i . Clearly, f_i lies inside a face of C . By Lemma 4.2.1 (ii), one of these is f . Since f_i meets f (close to x), we thus have $f_i \subseteq f$. Moreover, the plane edges xx_j with $j \notin \{i, i+1\}$ meet C_i only in x and end outside f_i in $C \setminus P_i$, so f_i meets none of those edges. Hence $f_i \subseteq \mathbb{R}^2 \setminus \tilde{G}'$, that is, f_i is contained in (and hence equal to) a face of $\tilde{G}'$. We may therefore extend $\tilde{G}'$ to a drawing of G by placing y and its incident edges in f_i . $\square$

 f_i

Compared with other proofs of Kuratowski's theorem, the above proof has the attractive feature that it can easily be adapted to produce a drawing in which every inner face is convex (exercise); in particular, every edge can be drawn straight. Note that 3-connectedness is essential here: a 2-connected planar graph need not have a drawing with all inner faces convex (example?), although it always has a straight-line drawing (Exercise 15).

It is not difficult, in principle, to reduce the general Kuratowski theorem to the 3-connected case by manipulating and combining partial drawings assumed to exist by induction. For example, if $\kappa(G) = 2$ and $G = G_1 \cup G_2$ with $V(G_1 \cap G_2) = \{x, y\}$, and if G has no TK^5 or $TK_{3,3}$ subgraph, then neither $G_1 + xy$ nor $G_2 + xy$ has such a subgraph, and we may try to combine drawings of these graphs to one of $G + xy$. (If xy is already an edge of G , the same can be done with G_1 and G_2 .) For $\kappa(G) \leq 1$, things become even simpler. However, the geometric operations involved require some cumbersome shifting and scaling, even if all the plane edges occurring are assumed to be straight.

The following more combinatorial route offers an ingenious alternative. In order to show that a given graph $G \not\supseteq TK^5, TK_{3,3}$ is planar we start by adding edges to G until it is edge-maximal with the property of not containing a TK^5 or $TK_{3,3}$. In Lemma 4.4.5 we show that this makes our graph 3-connected, so by Lemma 4.4.3 it is planar.

For the proof of Lemma 4.4.5 we need another lemma. We state this a little more generally, so that we can use it again in another context in Chapter 7. For our application here put $\mathcal{X} := \{K^5, K_{3,3}\}$.

Lemma 4.4.4. *Let $\mathcal{X}$ be a set of 3-connected graphs. Let G be a graph with a proper separation $\{V_1, V_2\}$ of order $\kappa(G) \leq 2$. If G is edge-maximal without a topological minor in $\mathcal{X}$, then so are $G_1 := G[V_1]$ and $G_2 := G[V_2]$, and $G_1 \cap G_2 = K^2$.*

[7.3.1]

Proof. Note first that every vertex $v \in S := V_1 \cap V_2$ has a neighbour in every component of $G_i - S$, $i = 1, 2$: otherwise $S \setminus \{v\}$ would separate G , contradicting $|S| = \kappa(G)$. By the maximality of G , every edge e added to G lies in a $TX \subseteq G + e$ with $X \in \mathcal{X}$. For all the choices of e considered below, the 3-connectedness of X will imply that the branch vertices of this TX all lie in the same V_i , say in V_1 . (The position of e will always be symmetrical with respect to V_1 and V_2 , so this assumption entails no loss

 S X

P

of generality.) Then the TX meets V_2 at most in a path P corresponding to an edge of X .

If $S = \emptyset$, we obtain an immediate contradiction by choosing e with one end in V_1 and the other in V_2 . If $S = \{v\}$ is a singleton, let e join a neighbour v_1 of v in $V_1 \setminus S$ to a neighbour v_2 of v in $V_2 \setminus S$ (Fig. 4.4.3). Then P contains both v and the edge $e = v_1v_2$; replacing its segment vPv_2v_1 with the edge vv_1 we obtain a TX in $G_1 \subseteq G$, a contradiction.

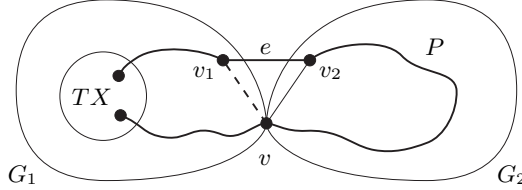


Fig. 4.4.3. If $G + e$ contains a TX , then so does G_1 or G_2

 x, y

So $|S| = 2$, say $S = \{x, y\}$. If $xy \notin G$, we let $e := xy$, and in the arising TX replace e by an x - y path through G_2 . This yields a TX in G , a contradiction. Hence $xy \in G$, and $G[S] = K^2$ as claimed.

It remains to show that G_1 and G_2 are edge-maximal without a topological minor in $\mathcal{X}$. So let e' be an additional edge for G_1 , say. Replacing xPy with the edge xy if necessary, we obtain a TX either in $G_1 + e'$ (which shows the edge-maximality of G_1 , as desired) or in G_2 (which contradicts $G_2 \subseteq G$). $\square$

Lemma 4.4.5. *If $|G| \geq 4$ and G is edge-maximal with $TK^5, TK_{3,3} \not\subseteq G$, then G is 3-connected.*

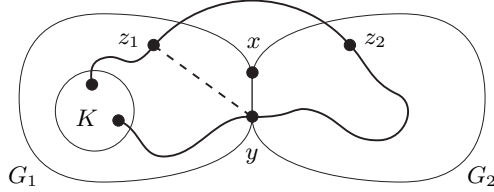
(4.2.11)

 G_1, G_2 x, y f_i z_i K

Proof. We apply induction on $|G|$. For $|G| = 4$, we have $G = K^4$ and the assertion holds. Now let $|G| > 4$, and let G be edge-maximal without a TK^5 or $TK_{3,3}$. Suppose $\kappa(G) \leq 2$, and choose G_1 and G_2 as in Lemma 4.4.4. For $\mathcal{X} := \{K^5, K_{3,3}\}$, the lemma says that $G_1 \cap G_2$ is a K^2 , with vertices x, y say, and that G_1 and G_2 too are edge-maximal without a TK^5 or $TK_{3,3}$. Hence, G_1 and G_2 are either a triangle or 3-connected by the induction hypothesis. Since they cannot contain K^5 or $K_{3,3}$ even as an ordinary minor (Lemma 4.4.2), they are thus planar by Lemma 4.4.3.

For each $i = 1, 2$ separately, choose a drawing of G_i , a face f_i with the edge xy on its boundary, and a vertex $z_i \neq x, y$ on the boundary of f_i . Let K be a TK^5 or $TK_{3,3}$ in the abstract graph $G + z_1z_2$ (Fig. 4.4.4).

If all the branch vertices of K lie in the same G_i , then either $G_i + xz_i$ or $G_i + yz_i$ (or G_i itself, if z_i is already adjacent to x or y , respectively) contains a TK^5 or $TK_{3,3}$; this contradicts Corollary 4.2.11, since these graphs are planar by the choice of z_i . Since $G + z_1z_2$ does not contain four independent paths between $(G_1 - G_2)$ and $(G_2 - G_1)$, these subgraphs cannot both contain a branch vertex of a TK^5 , and cannot both contain

Fig. 4.4.4. A TK^5 or $TK_{3,3}$ in $G + z_1z_2$

two branch vertices of a $TK_{3,3}$. Hence K is a $TK_{3,3}$ with only one branch vertex v in, say, $G_2 - G_1$. But then also the graph $G_1 + v + \{vx, vy, vz_1\}$, which is planar by the choice of z_1 , contains a $TK_{3,3}$. This contradicts Corollary 4.2.11. $\square$

Theorem 4.4.6. (Kuratowski 1930; Wagner 1937)

The following assertions are equivalent for graphs G :

- (i) G is planar;
- (ii) G contains neither K^5 nor $K_{3,3}$ as a minor;
- (iii) G contains neither K^5 nor $K_{3,3}$ as a topological minor.

Proof. Combine Corollary 4.2.11 with Lemmas 4.4.2, 4.4.3 and 4.4.5. $\square$

[4.5.1]
[5.1.2]
[12.6.4]

(4.2.11)

Corollary 4.4.7. Every maximal planar graph with at least four vertices is 3-connected.

Proof. Apply Lemma 4.4.5 and Theorem 4.4.6. $\square$

4.5 Algebraic planarity criteria

One of the most conspicuous features of a plane graph G are its *facial cycles*, the cycles that bound a face. If G is 2-connected it is covered by its facial cycles, so in a sense these form a ‘large’ set. In fact, the set of facial cycles is large even in the sense that they generate the entire cycle space: every cycle in G is easily seen to be the sum of the facial cycles (see below). On the other hand, the facial cycles only cover G ‘thinly’, as every edge lies on at most two of them. Our first aim in this section is to show that the existence of such a large yet thinly spread family of cycles is not only a conspicuous feature of planarity but lies at its very heart: it characterizes it.

facial
cycles

Let $G = (V, E)$ be any graph. We call a subset $\mathcal{F}$ of its edge space $\mathcal{E}(G)$ *sparse* if every edge of G lies in at most two sets of $\mathcal{F}$. For example, the cut space $\mathcal{B}(G)$ has a sparse basis consisting of atomic cuts (Proposition 1.9.2).

sparse